KAIZEN MATHS

Lesson starter

Surds and Radicals

Advanced Surd Techniques

Name _____

S1

Rationalise: $\frac{6}{\sqrt{2}}$

S2

Rationalise: $\frac{2}{6+\sqrt{2}}$

S3

Expand and simplify: (√7 - 5)(√7 - 3)

S4

Simplify using difference of two squares: (√3 + 4)(√3 - 4)

Show your working

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Exit ticket

Surds and Radicals

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Name _____

E1

Simplify: $4\sqrt{2} \times 3\sqrt{2} - 4\sqrt{8} + \sqrt{2}$

E2

Rationalise: $\frac{12}{\sqrt{7}}$

E3

Rationalise: $\frac{7}{5+\sqrt{2}}$

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Teacher answers & working

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S1

Rationalise: $\frac{6}{\sqrt{2}}$

Answer:

$3\sqrt{2}$

Working:

Multiply by $\frac{\sqrt{2}}{\sqrt{2}}$

$\frac{6}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$

Multiply numerator and denominator

$\frac{6\sqrt{2}}{2}$

Simplify the fraction if possible

$3\sqrt{2}$

S2

Rationalise: $\frac{2}{6+\sqrt{2}}$

Answer:

$\frac{6-\sqrt{2}}{17}$

Working:

Identify the conjugate

The conjugate of 6 + √2 is 6 - √2

Multiply by the conjugate over itself

$\frac{2}{6+\sqrt{2}} \times \frac{6-\sqrt{2}}{6-\sqrt{2}}$

Use difference of squares in the denominator

$A = 6$ and $B = \sqrt{2}$.

$A + B = 6 + \sqrt{2}$.

$A - B = 6 - \sqrt{2}$.

$6^2 - (\sqrt{2})^2 = 36 - 2 = 34$

Simplify the fraction

$\frac{6-\sqrt{2}}{17}$

Teacher answers & working

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S3

Expand and simplify: (√7 - 5)(√7 - 3)

Answer:

22 - 8√7

Working:

Complete the multiplication table

×	√7	- 3
√7	√7 × √7 = 7	√7 × (-3) = -3√7
- 5	(-5) × √7 = -5√7	(-5) × (-3) = 15

Find the pairs that can be simplified

Constant pair: 7 + 15 = 22

Like-surd pair: -3√7 - 5√7 = -8√7

Write the simplified expression

22 - 8√7

S4

Simplify using difference of two squares: (√3 + 4)(√3 - 4)

Answer:

-13

Working:

Recognise the conjugate pair

The brackets have the form (A + B)(A - B).

Use the difference of two squares identity

$(A + B)(A - B) = A^2 - B^2$.

Identify A and B

$A = \sqrt{3}$ and $B = 4$.

Show the two bracket factors

$A + B = \sqrt{3} + 4$.

$A - B = \sqrt{3} - 4$.

Square each term

$(\sqrt{3})^2 = 1^2 \times 3 = 3$, and $4^2 = 16$

Subtract the squares

3 - 16 = -13

-13

Teacher answers & working

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E1

Simplify: $4\sqrt{2} \times 3\sqrt{2} - 4\sqrt{8} + \sqrt{2}$

Answer:

$24 - 7\sqrt{2}$

Working:

Apply order of operations: multiply first

$4\sqrt{2} \times 3\sqrt{2} = 12 \times (\sqrt{2} \times \sqrt{2}) = 12 \times 2 = 24$

Simplify the remaining larger surd

$\sqrt{8} = 2\sqrt{2}$

Rewrite the expression

$24 - 8\sqrt{2} + \sqrt{2}$

Combine like surds

$-8\sqrt{2} + \sqrt{2} = -7\sqrt{2}$

$24 - 7\sqrt{2}$

E2

Rationalise: $\frac{12}{\sqrt{7}}$

Answer:

$\frac{12\sqrt{7}}{7}$

Working:

Multiply by $\frac{\sqrt{7}}{\sqrt{7}}$

$\frac{12}{\sqrt{7}} \times \frac{\sqrt{7}}{\sqrt{7}}$

Multiply numerator and denominator

$\frac{12\sqrt{7}}{7}$

Simplify the fraction if possible

$\frac{12\sqrt{7}}{7}$

Teacher answers & working

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E3

Rationalise: $\frac{7}{5+\sqrt{2}}$

Answer:

$\frac{7(5-\sqrt{2})}{23}$

Working:

Identify the conjugate

The conjugate of 5 + √2 is 5 - √2

Multiply by the conjugate over itself

$\frac{7}{5+\sqrt{2}} \times \frac{5-\sqrt{2}}{5-\sqrt{2}}$

Use difference of squares in the denominator

$A = 5$ and $B = \sqrt{2}$.

$A + B = 5 + \sqrt{2}$.

A - B = 5 - √2.

$5^2 - (\sqrt{2})^2 = 25 - 2 = 23$

Simplify the fraction

$\frac{7(5-\sqrt{2})}{23}$